

Annie Prasanna Manoharan

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EDUCATION

University of Arkansas, Fayetteville, AR

Aug 2024 – May 2026

Master of Science in Computer Engineering

Coursework: Algorithms, Database Management Systems, Advanced DBMS, Data Mining, Artificial Intelligence, Machine Learning, Image Processing, Cryptography, Privacy Enhancing Techniques

TECHNICAL SKILLS

Programming Languages: Python, SQL, Java, C++, C, C#

Software Engineering: Object-Oriented Programming, Data Structures & Algorithms, Software Design, Design Patterns, Multithreading, Concurrency, Exception Handling, Debugging, Troubleshooting, Code Review, SDLC

Machine Learning: PyTorch, Scikit-learn, NumPy, Pandas, Deep Learning, Machine Learning, Neural Networks, Computer Vision, Object Detection, Object Tracking, Image Processing, Predictive Modeling, Model Training, Model Validation

Generative AI: LLMs, RAG, RAG Pipelines, LLM Integration, Vector Embeddings, Semantic Search, LangChain, FAISS, Ollama

Data Engineering: ETL, ELT, Data Pipelines, Data Ingestion, Data Extraction, Data Transformation, Data Validation, Data Cleaning, Data Processing, Batch Processing, Real-Time Data Processing, Workflow Automation

Databases: SQL Server, PostgreSQL, MySQL, Oracle, MongoDB, ClickHouse, Relational Databases, NoSQL, Database Design

Cloud & Tools: AWS, Docker, Apache Airflow, Apache Kafka, Apache Flink, Git, CI/CD, Agile, Scrum

PROFESSIONAL EXPERIENCE

Research Assistant, AICV Lab, University of Arkansas

Sep 2024 – Jan 2026

Project: [MiGa: Multi-Chicken Gait Assessment](#)

- Integrated YOLO11n-Pose for keypoint estimation, applied OC-SORT for multi-object tracking (MOT), and constructed a many-to-one LSTM sequence network for classification
- Built a data engineering framework to preprocess and segment raw frames into a Multi-Bird Detection split of 4,569 frames with 11,422 bounding boxes and a Gait-Score Regression split spanning 55 expert-annotated sequences
- Implemented spatial normalization to map keypoints into local coordinate spaces to eliminate translational variance, achieving a keypoint mAP of 91.83% at 1,250 FPS alongside 0.93 MOTA and 0.91 IDF1 tracking accuracy

Project: [DualFit: Two-Stage Virtual Try-On via Warping and Synthesis](#)

- Developed a PyTorch-based two-stage Virtual Try-On (VTON) model combining flow-based garment warping with Res-UNet image synthesis, with preprocessing and inpainting pipelines to improve garment alignment and visual realism
- Trained and evaluated the model on the VITON-HD dataset, achieving 24.9 PSNR, 0.913 SSIM, and 5.261 FID

Project Engineer, Soliton Technologies, India

July 2022 – June 2024

- Designed and developed a C# WPF desktop application integrating C#, Python, and LabVIEW-based test workflows to automate testing and streamline application usage
- Implemented reusable, object-oriented software components and automated workflows, improving development efficiency, test coverage, and repeatability
- Served as the sole developer for a customized Medtronic solution, automating approximately 94% of manual test cases while meeting medical-device quality and validation requirements
- Collaborated with cross-functional engineering teams to debug system-level issues, troubleshoot application behavior, optimize workflows, and deliver production-ready software solutions

PROJECTS

Real-Time Medical IoT Analytics Platform

Apache Kafka, Apache Flink, SQL, Python, ClickHouse, Grafana

- Developed an end-to-end, sub-second latency streaming platform using Apache Flink SQL and Apache Kafka to ingest, process, and score high-frequency medical IoT device data
- Implemented temporal Flink OVER windows partitioned by device ID to dynamically compute rolling Z-scores over 5-second sliding intervals, applying custom watermark delay strategies to cleanly handle out-of-order data streams
- Designed high-performance ClickHouse schemas using the MergeTree engine sorted by composite keys to physically align disk layouts with clinical querying patterns, while isolating ingestion via an optimized Python multi-topic batch consumer
- Integrated a Grafana visualization layer directly on top of the ClickHouse OLAP tier, delivering real-time clinical dashboards that monitor raw ventilator telemetry and instantly surface streaming Z-score anomalies

Expense Tracker with Predictive Analytics

C++, MySQL, Python, SQL, Database Integration

- Designed and implemented a cross-language expense management application integrating C++ business logic with a MySQL database for storing, retrieving, categorizing, and processing historical expense data
- Developed and optimized SQL queries to generate category-based expense summaries and reports, integrating database operations with the application's business logic
- Integrated Python-based analytics with the application workflow to analyze historical spending patterns and generate next-month expense predictions